

Particulate Matter Monitoring & Health Effects

Issue 2026-09-18 | Entry window 18 September 2026 (full day) | 30 records screened in scope

30RECORDS IN SCOPE
(102 EXCLUDED)**8**OF 10 SUBTOPIC
BANDS POPULATED**7**EFFECT ESTIMATES
(ONLY 3 WITH A CI)**5**TIER-A
RECORDS**8**RECORDS CARRIED ON
METADATA ALONE

Signal of the day

Four independent records today attack the same assumption from four directions: that a surface-sited, mass-reporting monitor measures the exposure that matters. Source, height, composition and speciated dose each break it.

Source. Zhang et al. (*Nat Commun*) separated wildfire-specific from non-wildfire PM_{2.5} across **20 US states, 2006–19**, in a self-controlled design. Per 1 µg/m³, wildfire PM_{2.5} raised cardiopulmonary hospitalisation risk **10% (heart failure) to 16% (asthma)**; non-wildfire PM_{2.5} raised it **4.7% (COPD) to 8.5% (hypertension)**. Equal mass, unequal hazard, by roughly a factor of two. **Height.** Bai et al. (*Atmos Meas Tech*) used Doppler wind lidar over Hefei to show two dust episodes with opposite surface signatures: in one, a descending low-level jet drove aerosol down and surface PM₁₀ peaked at **410 µg/m³**; in the other, a stable jet core held the layer aloft and the surface peak arrived **~8 h** after frontal onset. A surface network reads clean air while a loaded layer sits overhead. **Composition.** Yu et al. (*Atmos Chem Phys*) found marine organic-aerosol light absorption over the Yellow and Bohai Seas rising with both organic-nitrogen concentration and ON/OC ratio — absorption capability per unit carbon changes with continental outflow, which is precisely the term a fixed optical calibration assumes constant. **Speciated dose.** Morrel et al. (*Biol Psychiatry Glob Open Sci*), the largest human study in the issue at **6,949 ABCD preadolescents**, finds the signal in *components* and not in mass: zinc, ammonium and bromine predicted perivascular-space count; calcium, zinc and ammonium predicted poorer cognition (FDR $p < 0.01$); and frontal-lobe perivascular-space count **mediated** the zinc-cognition association. Total PM_{2.5} entered only as a weighted-quantile-sum mixture, and never as the quantity a monitor reports. Only one of the four is a sensing paper. Together they are the strongest argument this watch has carried that the next increment in exposure assessment is source-, height- and composition-resolved, not denser mass.

What changed today

- ▶ **A three-year reference-grade absorption record at a rural South Asian site.** Raju et al. (*J Environ Sci Health A*) report seven-wavelength AE33 aethalometry at Panchgaon, Haryana, **January 2020 – December 2022**: eBC **4.06 ± 2.15**, fossil-fuel **3.10 ± 2.40** and biomass-burning **1.42 ± 1.19 µg/m³**, with eBCff carrying **68%** of absorption at 880 nm and winter maxima roughly double the monsoon minima. This is the co-location anchor low-cost networks in the region do not have. The apportionment itself is shakier than the record: the eBCff/eBCbb ratio is **3.02 ± 3.80** — a standard deviation larger than the mean — and the assumed absorption Ångström exponents are never stated.
- ▶ **A community-designed monitoring campaign that is a template, not a result.** Frueh et al. (*J Expo Sci Environ Epidemiol*) ran **38 VOCs at 20 sites for 12 months** in a South Philadelphia fenceline community with Philly Thrive, around a former refinery. Benzene — the pollutant the community named — came in low (**median 1.21 µg/m³**), while (*m,p*)-xylenes emerged in industrial areas, and PMF apportioned sources. One-week integrated passive samplers resolve none of the short plumes residents actually report, which is the design's own limit. The siting logic — community-set locations, a declared pollutant of concern, apportionment rather than compliance — is what distributed PM networks should copy.
- ▶ **A humidity threshold that lands squarely on the sensor-correction problem.** Sun et al. (*Environ Sci Technol*) split wintertime basin nitrate formation almost evenly between daytime OH+NO₂ (**52%**) and nighttime N₂O₅ hydrolysis (**46%**), and locate abrupt efficiency thresholds at **RH 87% and ALWC 76.5 µg/m³**. That is the same humidity band in which optical low-cost sensors diverge from reference monitors — the aerosol is physically changing where the instrument is least trustworthy. One site, one winter, no out-of-sample validation.
- ▶ **A cardiovascular case-crossover in which every PM term is null.** Chiang et al. (*J Am Heart Assoc*) found no same-day association for any pollutant across **798 acute aortic dissections**; only CO at 2-day lag survived (**OR 2.13 per IQR, 95% CI 1.075–4.239**). PM_{2.5}, PM₁₀, NO₂, O₃ and SO₂ showed nothing at any lag. With six pollutants across four lags and no multiplicity correction, the surviving association is fragile — but the PM nulls are themselves informative: exposure was assigned from **24 fixed stations**, a density at which within-city contrast is largely erased.

► **Eight of thirty records assert no finding at all.** Metadata-only is the largest single design bucket in this issue, ahead of measurement campaigns at seven. Same-day deposits from Elsevier, ACS, BMC and now Taylor & Francis carry a title, authors and a DOI and nothing else. A new recovery route was tested and **failed**: the OpenAlex per-work endpoint (`/works/doi:`), which is free even though the date-range filter is not, returns `abstract_inverted_index` only where the publisher deposited an abstract to Crossref in the first place — so it recovers nothing these publishers withhold.

Run note — this issue was rebuilt

A first build of this issue ran at **10:33 CDT** and shipped 17 records against roughly ten hours of a twenty-three-hour entry window, contrary to the 22:00 rule the pipeline adopted on 19 August. Re-harvesting the same window at **23:0x** returned **PubMed 12** (was 3), **Europe PMC 46** (was 44) and **Crossref-by-ISSN 53** (was 24); 13 of the 72 fresh candidates were admitted. This issue supersedes the morning build and covers the whole of 18 September. The earlier version was never the complete day, and no record it carried has been removed.

Corpus analytics

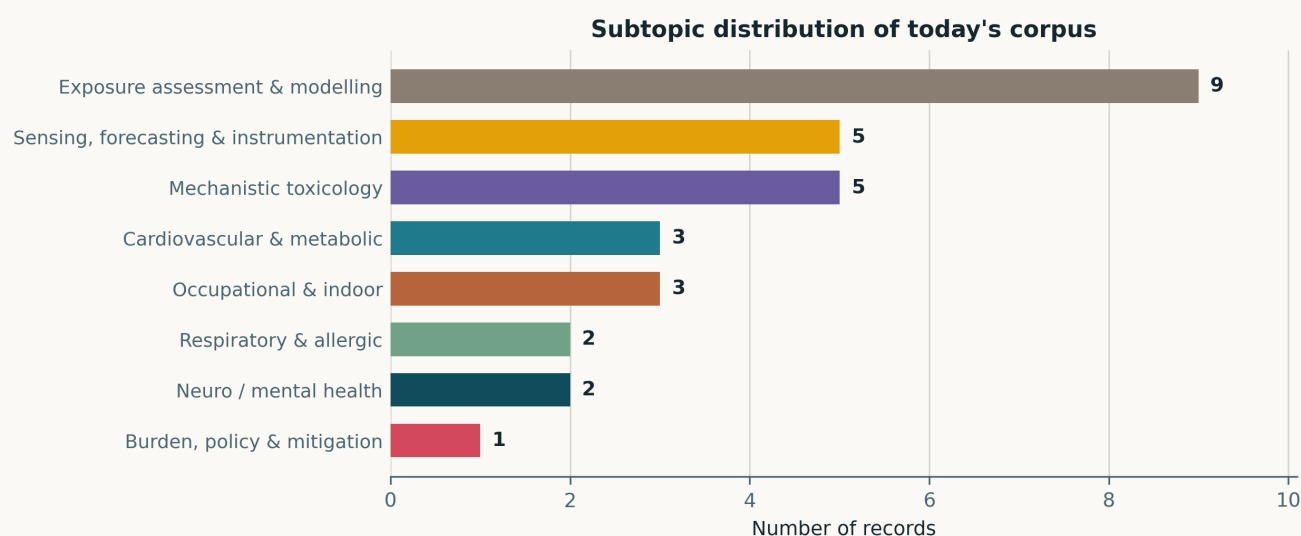


Fig. Subtopic assignment of the 30 in-scope records; eight of the ten bands are populated. *Exposure assessment & modelling* is the largest at nine, and with *Sensing, forecasting & instrumentation* at five the measurement side holds **14 of 30**. *Mechanistic toxicology* also reaches five, then *Cardiovascular & metabolic* and *Occupational & indoor* at three each, *Respiratory & allergic* and *Neuro / mental health* at two, and *Burden, policy & mitigation* at one. *Reproductive & developmental* and *Other clinical endpoints* are empty.

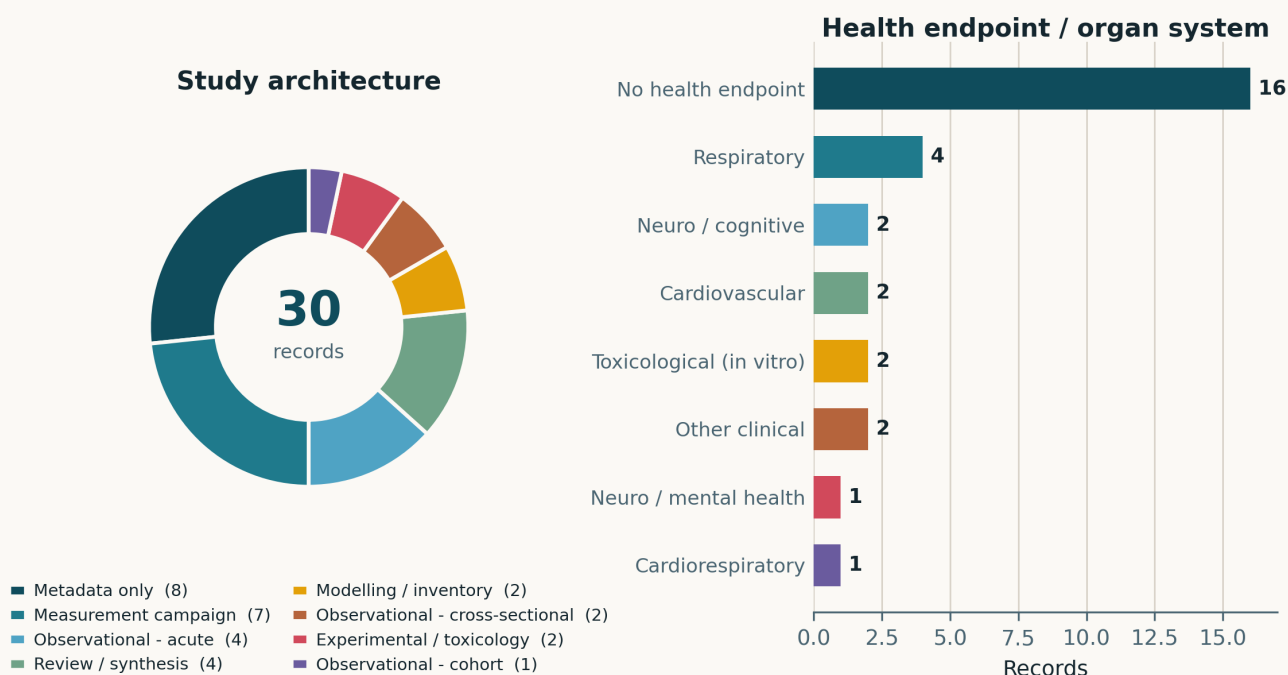


Fig. Left — study architecture. **Metadata only is the largest single bucket at eight of thirty**, ahead of measurement campaigns at seven; that ordering, not the total, is the finding, and it is a property of same-day publisher deposits rather than of the literature. Primary observational epidemiology contributes seven (four acute, two cross-sectional, one cohort), evidence synthesis four, modelling two and experimental toxicology two. Right — health endpoint, canonicalised. **“No**

health endpoint” is 16 of 30 — the aerosol-chemistry, emissions, source-policy and not-applicable records all collapse into it — so **more than half the issue reports no human outcome**. Respiratory leads the clinical endpoints at four; cardiovascular, neuro / cognitive, other clinical and toxicological (in vitro) carry two each.

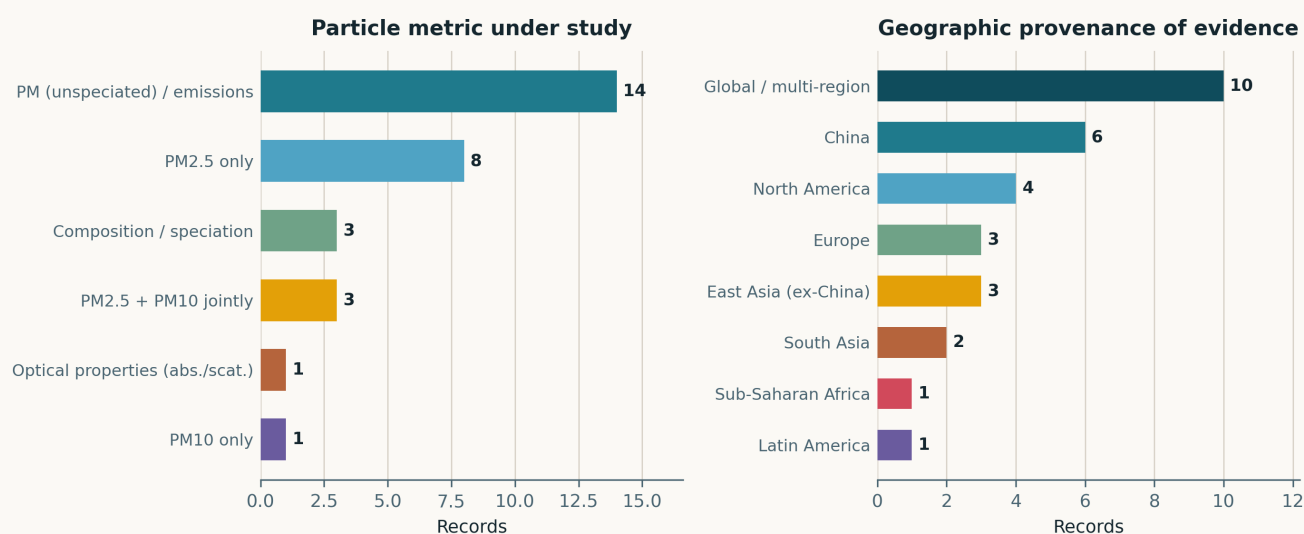


Fig. Left — particle metric. **Unspeciated PM or emissions terms cover 14 of 30**, and the composition of that bar is the point: it absorbs both the metadata-only records and the four VOC / precursor studies, none of which measured a particle mass at all. PM_{2.5} alone accounts for eight, speciated composition and joint PM_{2.5} + PM₁₀ three each, PM₁₀ alone and optical properties one each. Right — geographic provenance. The *Global / multi-region* bar is ten and again is not what it looks like: **four are records whose study location is not recoverable**, and of the remaining six, three are reviews or commentaries with no field site. China is the largest genuine single-country bar at six, then the USA at four.

Life-course windows addressed today (records may span several stages)

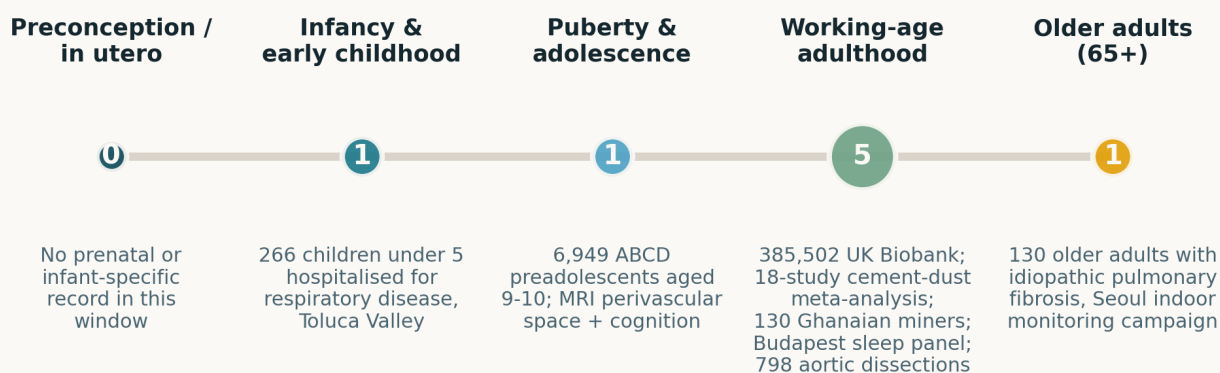


Fig. Life-course windows addressed; counts are non-exclusive. Working-age adulthood carries five — the UK Biobank COPD cohort, the 18-study cement-dust worker meta-analysis, the Ghanaian mining communities, the Budapest sleep panel and the 798 aortic-dissection cases. Puberty and adolescence is newly occupied this issue by the **6,949 ABCD preadolescents**, and older adults by the 130 Seoul patients with idiopathic pulmonary fibrosis. The paediatric entry is the Toluca series of 266 children under five; because bronchiolitis cases there are necessarily infants but no infant stratum is tabulated, the prenatal/infant window is recorded as zero rather than inferred.

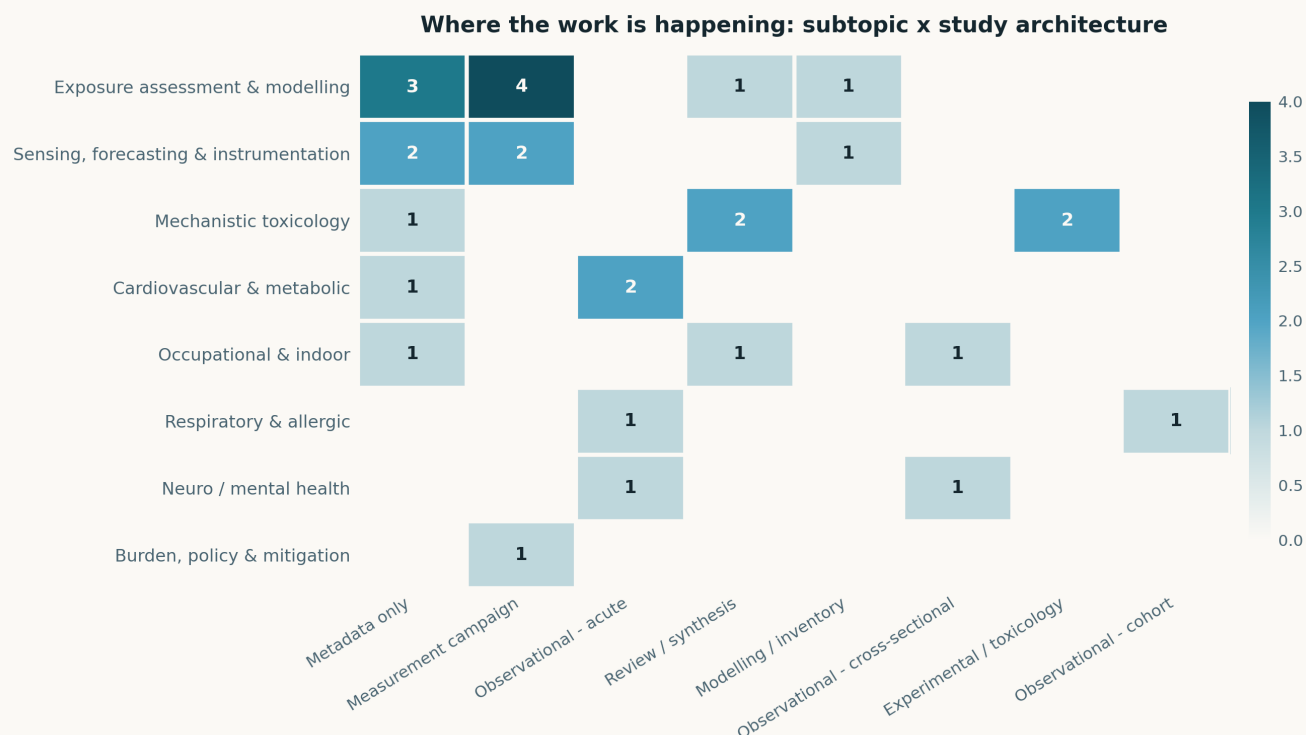


Fig. Subtopic × study-architecture cross-tabulation. The structural fact is unchanged by tripling the sensing row and doubling the modelling row: **the Exposure assessment & modelling row (nine records) and the Sensing row (five) canonicalise to “no health endpoint” without a single exception.** Fourteen records — the entire measurement side of the issue — report no human outcome, and no record anywhere in the matrix links an instrument to a clinical endpoint. The second feature is dispersion: metadata-only entries appear in six of the eight populated rows, so the abstract-free deposit problem is not confined to one literature.

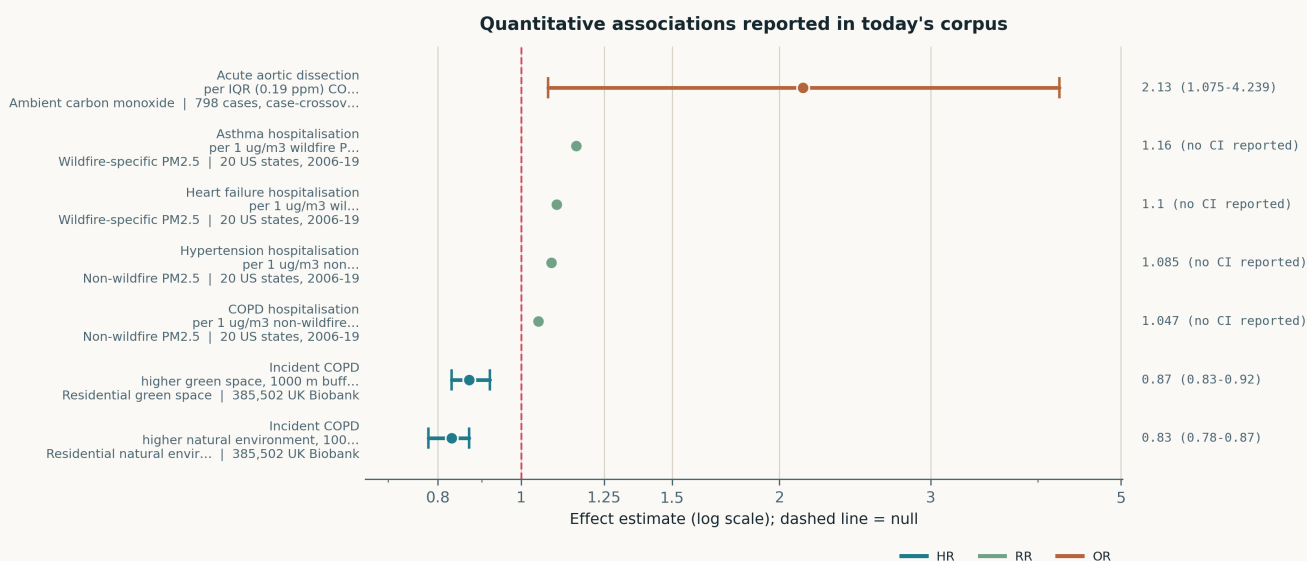


Fig. All seven extractable effect estimates on a common logarithmic axis; only three report a confidence interval. Those three now bracket the null in both directions — two protective hazard ratios from the UK Biobank greenness analysis, and one **OR 2.13 (95% CI 1.075–4.239)** for acute aortic dissection per IQR of CO at 2-day lag — while **all four estimates above the null that come from PM exposure are point values with no interval reported.** Exposure contrasts are not mutually comparable: hazard ratios per greenness category at a 1000 m buffer, risk ratios per 1 $\mu\text{g}/\text{m}^3$ of source-resolved PM_{2.5}, an odds ratio per 0.19 ppm of CO. Read direction and precision, never relative magnitude.

Paper-level digest

Ordered by cluster. Each entry gives the methodological core and the finding that matters, not an abstract paraphrase. Records marked *metadata only* carry no recoverable abstract and assert no finding.

Neuro, cognition & mental health

2 records

Temporao et al. 2026 medRxiv (preprint)

Daily associations between outdoor air pollution and sleep quality

Budapest Sleep, Traits and Experiences Study — seven days per participant of daily activity and sleep-quality reports plus home EEG giving sleep efficiency, %REM, onset latency and total sleep time. Linear mixed models with participant random intercepts, adjusted for age, sex, BMI, education, weekday, season, temperature, physical activity, screen exposure and travel. **Day-1 ozone predicted lower day-2 sleep efficiency and poorer perceived quality**, surviving multiple-comparison correction, multi-pollutant models and sensitivity analyses for depression, insomnia and chronotype; **no robust association for PM_{2.5} or NO₂**. The null is confounded with exposure-model quality — city-aggregated station data give near-zero within-city contrast, and misclassification is worse for PM_{2.5} than for the more spatially homogeneous ozone. Sample size is not stated in the abstract.

doi: 10.64898/2026.09.15.26363130

Morrel et al. 2026 Biol Psychiatry Glob Open Sci

Outdoor air pollution, perivascular space morphology, and cognition in preadolescence

Cross-sectional analysis of **6,949 ABCD participants aged 9–10**, with annual-average PM_{2.5}, O₃, NO₂ and **15 PM_{2.5} components** assigned to residential addresses by spatiotemporal models, perivascular-space (PVS) count and volume from T1- and T2-weighted MRI, and cognition from the NIH Toolbox. **Zinc, ammonium and bromine** were positively associated with regional PVS count; higher count in five regions tracked poorer cognition; **calcium, Zn and NH₄⁺** were associated with poorer cognition (FDR-corrected $p < 0.01$); frontal-lobe PVS count **mediated** the Zn–total-cognition association ($p < 0.01$); and weighted-quantile-sum PM_{2.5} mixtures predicted higher temporal and cingulate PVS with poorer working memory and crystallised intelligence. The design cannot order exposure, PVS and cognition, and the abstract reports significance without a single effect size. The component-specific pattern is the measurement claim: speciation, not mass, carries the signal.

doi: 10.1016/j.bpsgos.2026.100790

Cardiovascular & metabolic

3 records

Zhang et al. 2026 Nat Commun

Cardiopulmonary hospitalisation risks from wildfire-specific and non-wildfire PM_{2.5} in 20 US states

Self-controlled design over 20 states, 2006–19, contrasting two-year average wildfire-specific against non-wildfire PM_{2.5}. Per 1 µg/m³, wildfire PM_{2.5} carried **10% (heart failure) to 16% (asthma)** higher hospitalisation risk; non-wildfire PM_{2.5} **4.7% (COPD) to 8.5% (hypertension)**. Associations were stronger among minorities, metropolitan residents, the less educated and more deprived communities. The toxicity contrast at equal mass is the contribution, and it undercuts the mass-only metric every regulatory network and low-cost sensor reports. Two limits: no confidence interval is given for any of the eight headline risks, so precision cannot be judged; and wildfire-specific PM_{2.5} is itself modelled, so the toxicity difference is conditional on a smoke-attribution step whose error is not propagated.

doi: 10.1038/s41467-026-76974-7

Chiang et al. 2026 J Am Heart Assoc

Ambient carbon monoxide as a delayed trigger of acute aortic dissection: a time-stratified case-crossover study in a tropical industrial metropolis

Time-stratified case-crossover of **798 emergency-department-confirmed acute aortic dissections**, 2018–2021, exposures assigned from **24 monitoring stations**, conditional logistic regression across lags 0–3 adjusted for meteorology. **No pollutant was associated on the event day**. At 2-day lag each 1 ppm CO increment carried a 2.67-fold risk (95% CI 1.207–5.900) and each IQR increment of 0.19 ppm raised risk **113% (95% CI 1.075–4.239)**; PM_{2.5}, PM₁₀, NO₂, O₃ and SO₂ **showed nothing at any lag**. Two problems: a 1 ppm contrast is roughly five IQRs, so the headline number extrapolates past the observed range; and with six pollutants across four lags the one surviving association is not multiplicity-corrected. The PM nulls are the relevant part here — station-assigned exposure at this network density returns nulls that a denser deployment might resolve.

doi: 10.1161/jaha.126.049936

Fleming & Fleming 2026 — *metadata only* Am J Med

Tire wear pollution and cardiovascular risk

Elsevier deposited title and authorship on 18 September with no abstract in Crossref, Europe PMC core, Europe PMC fullTextXML or the OpenAlex per-work record. The title places it as a commentary linking tire-wear particle pollution to cardiovascular risk; it asserts no finding here. Logged because the particle class matters for measurement: **non-exhaust emissions now dominate traffic PM in fleets with modern aftertreatment**, and tire wear differs from combustion PM in both size distribution and composition — a combination that mass-only optical sensors and exhaust-derived emission factors each mis-handle. Carried forward for pickup when an abstract appears.

doi: 10.1016/j.amjmed.2026.09.003

Respiratory & allergic

2 records

San Juan-Reyes et al. 2026 Environ Sci Pollut Res Int

Short-term ambient air pollution and respiratory hospitalisation in children under five

266 children under five admitted to one regional IMSS hospital in the Toluca Valley Metropolitan Zone during 2023; matched case-crossover with conditional logistic regression, primary window lag 1–7, adjusted for temperature and humidity. PM_{2.5} was associated with admission for viral pneumonia, bacterial pneumonia and asthma, CO with bronchiolitis, with heterogeneous municipality-level associations. **The abstract reports direction only — not one odds ratio, interval or unit appears** — so the effect size is unverifiable from the published summary. 266 events split across four diagnoses, five pollutants, several lag windows and three municipalities is thin, and exposure is entirely fixed-station with no personal or indoor component in a setting where indoor sources plausibly dominate a young child's dose.

doi: 10.1007/s11356-026-38237-2

Chen et al. 2026 Int J Chron Obstruct Pulmon Dis*Residential green and blue space, genetic susceptibility and incident COPD*

385,502 UK Biobank participants, Cox models, greenness at 300 m and 1000 m buffers against a COPD polygenic risk score. At 1000 m: **green space HR 0.87 (95% CI 0.83–0.92)**, **natural environment HR 0.83 (0.78–0.87)**; blue space weak and inconsistent. No multiplicative or additive interaction with genetic risk, and the natural-environment association held in every genetic stratum — so the protective signal is not running through a susceptible subgroup. The inferential problem is what is absent: **no air-pollution term is reported**. Residential greenness is inversely correlated with traffic PM_{2.5} and NO₂ nearly everywhere, so an unadjusted greenness–COPD association may simply restate a pollution–COPD association.

doi: 10.2147/copd.s627458

Mechanistic toxicology**5 records****Gao et al. 2026** Front Public Health*CCL22-associated immune signatures in air pollution-related atopic dermatitis*

Shared pollutant–atopic dermatitis targets enriched in cytokine and chemokine signalling, pattern-recognition receptor activity, IL-17 and Toll-like receptor pathways. A framework over 15 algorithms and 175 combinations selected plsRglm + AdaBoost, with AUC 0.963 in training and 1.000, 0.909 and 0.966 across three validation cohorts, giving a ten-gene signature (CCL22, CCL5, CSF2, F2RL1, HRH4, ICAM1, IFNG, IL10, IL17A, IL18). CCL22 was upregulated in lesional samples, localised to dendritic cells and macrophages, and confirmed in DNFB-induced murine dermatitis. **The exposure link is the weak joint: no particulate matter was measured or administered anywhere**, so air pollution enters only as a database annotation. An AUC of exactly 1.000 indicates leakage or overfitting rather than performance, and DNFB is a contact hapten, not a particle.

doi: 10.3389/fpubh.2026.1906298

Vera-Colón et al. 2026 — metadata only Part Fibre Toxicol*Airborne fine particulate matter and human embryonic stem cell osteogenic differentiation*

BMC deposited this DOI on 18 September with no abstract; Crossref, Europe PMC core, Europe PMC fullTextXML and the publisher landing page all returned nothing on per-DOI retry — **the first time this gap has appeared on a BMC rather than an Elsevier deposit**. Carried on title because skeletal development is an endpoint the PM literature has barely touched, and a human-embryonic-stem-cell osteogenic model would be a new mechanistic line rather than another airway or endothelial assay. No finding is asserted: neither particle source, dose, duration nor the direction of the reported disruption is verifiable from metadata. On the retry list.

doi: 10.1186/s12989-026-00701-9

Dulong et al. 2026 Biochimie*Effects of fine particulate matter on neuroblastoma cell line: metabolic and lipid deregulations and their consequences on the onset of neurological diseases*

Narrative review built around the **SH-SY5Y** human neuroblastoma line as an in vitro neurodegeneration model. The proposed cascade runs mitochondrial damage → ER stress → oxidative stress → calcium dyshomeostasis → lipid dysregulation, converging on apoptosis and autophagy-dependent **ferroptosis**, with lipid peroxidation advanced as the emerging driver. No pooled estimate, no exposure–response and no dose metric: the review does not harmonise PM_{2.5} mass, source or exposure duration across the studies it assembles, and the in vitro doses it draws on generally sit far above ambient, so the chain is qualitative. It concedes its own in-vitro-to-human translation gap. Every mechanism it privileges is **composition-dependent** — redox-active metals, organic carbon — which a mass-only network cannot resolve.

doi: 10.1016/j.biochi.2026.09.011

Wu et al. 2026 Ecotoxicol Environ Saf*β-carotene alleviates PM_{2.5}-induced pulmonary fibrosis via suppression of the Frizzled5-mediated Wnt5a/Ca²⁺ pathway*

C57BL/6 mice given **intratracheal PM_{2.5}**, paired with BEAS-2B human bronchial epithelial cells carrying Fzd5 overexpression. Instillation produced fibrosis, oxidative stress and calcium dyshomeostasis — more mitochondria-associated ER membranes, altered IP₃R, STIM1, ORAI1 and SERCA2 — with activation of a **Wnt5a/Fzd5/Ca²⁺** axis. Dietary β-carotene reversed these changes, and Fzd5 overexpression abolished the protection, identifying Fzd5 as the mediating receptor. **No instilled dose, particle source, treatment concentration, effect size or CI** is reported for any endpoint, so potency cannot be placed against ambient exposure. Intratracheal instillation also bypasses the deposition physics that a sensor-derived exposure metric is meant to approximate.

doi: 10.1016/j.ecoenv.2026.120816

Kou et al. 2026 Front Toxicol*Pollutants regulate changes in pathological markers of neurodegenerative diseases: a new perspective in environmental toxicology*

Review organised around core neurodegenerative pathological markers — **amyloid-β, tau, α-synuclein, TDP-43** — rather than around individual pollutants, detailing how air pollutants, metals and persistent organic pollutants modulate the generation, aggregation, propagation and toxicity of each, under a proposed “pollutants–pathogenic links–neural network failure” framework. No quantitative synthesis, no effect estimates, no exposure metric and no study-quality appraisal; it reorganises existing mechanistic claims rather than adding evidence, and **air pollutants are treated as a single undifferentiated class** alongside metals and POPs. The one useful implication for measurement is that marker-level convergence makes pollutant class, not PM mass, the exposure variable worth resolving.

doi: 10.3389/ftox.2026.1814795

Occupational & indoor exposure**3 records****Khan et al. 2026** Front Public Health*Cement industry emissions and respiratory health: global systematic review and meta-analysis, 2000–2025*

Eighteen observational studies, PROSPERO CRD420251268913. Random-effects pooling gave standardised mean differences of **–0.71 for FEV₁, –0.56 for FVC and –0.57 for FEV₁/FVC** in exposed groups; chronic cough was significantly more common while other symptoms were elevated but not significant. Occupational exposure produced greater FEV₁ impairment than residential — the expected dose ordering and the most reassuring internal check available. Reporting is the weakness: **no confidence interval, no I² and no prediction interval accompanies any pooled estimate**, so neither precision nor heterogeneity can be assessed, and SMDs pooled across studies using different spirometry reference equations with no common dust metric cannot be read as a dose–response.

doi: 10.3389/fpubh.2026.1887654

Assan et al. 2026 Health Sci Rep*Ambient particulate matter, urinary arsenic and resident health in two Ghanaian mining communities*

65 residents each from an artisanal and an incorporated mining community, all resident at least five years; PM by Modular Air Quality System, urinary arsenic by AAS, kidney function by albumin–creatinine ratio. More artisanal residents fell in the at-risk band for kidney disease ($p = 0.006$); the incorporated community showed poorer general health status. Descriptive only: $n = 130$, chi-square on dichotomised uACR with no confounder adjustment and no exposure–response, and **neither PM concentrations nor arsenic levels are numerically reported** despite being the two measured exposures. The 35 and 70 $\mu\text{g}/\text{m}^3$ figures quoted as WHO standards are the 2005-era interim targets, not the 2021 guideline values of 15 and 45 $\mu\text{g}/\text{m}^3$ for 24-hour $\text{PM}_{2.5}$ and PM_{10} — so the exceedance framing understates the gap.

doi: 10.1002/hsr2.73246

Ikeda et al. 2026 — metadata only Build Environ*Characterizing layout-dependent breathing-zone contaminant concentration in office spaces with mixing ventilation using an occupant-to-exhaust distance metric*

Same-day Elsevier deposit; nothing in Crossref, Europe PMC core, Europe PMC fullTextXML or the OpenAlex per-work record. The title indicates a modelling study deriving breathing-zone contaminant concentration in mixing-ventilated offices from an **occupant-to-exhaust distance metric**. In scope for sensor siting specifically: it is a claim that indoor concentration is a predictable function of geometry, which is the assumption every single-point indoor monitor makes implicitly and almost never tests. Carried forward for pickup.

doi: 10.1016/j.buildenv.2026.115282

Sensing, forecasting & instrumentation**5 records****Bai et al. 2026** Atmos Meas Tech*Low-level jet evolution and vertical aerosol redistribution: a Doppler wind lidar study*

Two dust episodes over Hefei, East China. April 2021: the northwesterly jet core descended from 2.5 km to 0.3 km then re-ascended to 1.0 km while surface PM_{10} peaked at 410 $\mu\text{g}/\text{m}^3$, with shear above 0.04 s^{-1} and increased spectral width at the lower jet interface and a downward shift of enhanced backscatter. March 2022: a stable jet core and persistent lower-tropospheric stability held aerosol aloft, the PM_{10} peak arriving at 16:00 LT, **~8 h after inferred frontal onset**. Two regimes — rapid downward mixing during jet descent versus delayed surface impact from a decoupled layer. Two cases is mechanism illustration, not climatology, and the inertial-oscillation attribution rests on hodograph rotation alone. The clearest statement yet of why a surface-only network can read clean for hours.

doi: 10.5194/amt-19-5951-2026

Huang et al. 2026 Environ Sci Technol*Interpretable spatiotemporal coupling network for emission prediction in sludge gasification*

PISTC-Net couples a directed process topology with per-feature optimal time lags, reaching R^2 of **91.18% for SO_2 , 80.37% for NO_x and 78.09% for PM** on industrial-scale data. Under simulated abrupt sensor failure it continued tracking emission trajectories via learned process inertia, and hierarchical attention shifted from upstream thermochemical zones for SO_2 to whole-process attention for PM, flagging disturbance propagation **5–8 minutes** before emission peaks. That PM is the worst-predicted species is the informative result: particle formation runs through nucleation, condensation and agglomeration paths a topology-plus-lag representation captures least well, whereas gas-phase sulfur is nearly a mass balance. Single plant, no external validation site, and interpretability is inferred from attention maps, which are a saliency artefact rather than a causal readout.

doi: 10.1021/acs.est.6c09135

Kalluri et al. 2026 — metadata only Atmos Environ*Long-term (2008–2024) black carbon over a semi-arid rain-shadow region in southern peninsular India*

The expected same-day Elsevier deposit gap: no abstract in Crossref, Europe PMC or PubMed on per-DOI retry. Carried on title because a 17-year single-site black-carbon record in a semi-arid rain-shadow setting is the kind of long baseline trend detection needs and South Asian networks mostly lack, and because absorption-coefficient records are what constrain the optical-to-mass conversion every filter- and scattering-based instrument depends on. No trend, magnitude or source attribution is asserted — the title names methods, not results. On the retry list.

doi: 10.1016/j.atmosenv.2026.122366

Krajnc Pogačnik et al. 2026 — metadata only Build Environ*Selective real-time monitoring of respiratory droplets for indoor air safety in educational environments*

Elsevier same-day deposit, abstract absent from Crossref, Europe PMC and PubMed on per-DOI retry. Retained because selective real-time discrimination of respiratory droplets against a background of ambient particles is the hard problem for any optical counter deployed indoors: a scattering sensor reports total number in a size bin and cannot by itself separate exhaled aerosol from resuspended dust, so any claimed selectivity mechanism is worth reading. Whether that selectivity is optical, chemical or inferential is not recoverable from the title, and no performance claim is asserted. On the retry list.

doi: 10.1016/j.buildenv.2026.115281

Raju et al. 2026 J Environ Sci Health A*Investigations of black carbon aerosol sources and strengths over a rural-valley location in Haryana, India*

Three years of seven-wavelength **AE33 aethalometry** (January 2020 – December 2022) at Panchgaon, a rural Aravali valley site at 285 m AMSL in northwest India. Annual means: eBC **4.06 ± 2.15**, fossil-fuel eBCff **3.10 ± 2.40**, biomass-burning eBCbb **1.42 ± 1.19 $\mu\text{g}/\text{m}^3$** , with eBCff carrying **68%** of absorption at 880 nm. Winter maxima (eBCff 4.19, eBCbb 2.51 $\mu\text{g}/\text{m}^3$) and monsoon minima (1.90, 0.52) track boundary-layer height and wet scavenging. The aethalometer-model split rests on **assumed absorption Ångström exponents that are never reported**, and the eBCff/eBCbb ratio of **3.02 ± 3.80** has a standard deviation larger than its mean — the apportionment is unstable even where the record is not. A three-year reference-grade absorption series at a rural South Asian site is exactly the co-location anchor low-cost networks there lack.

doi: 10.1080/10934529.2026.2734478

Exposure assessment & modelling

9 records

Kim et al. 2026 Environ Health*Residential indoor air monitoring for long-term exposure assessment in susceptible populations*

130 older adults with idiopathic pulmonary fibrosis, Seoul Metropolitan Area, 2021–23. Indoor hourly PM_{2.5} averaged **14.90 $\mu\text{g}/\text{m}^3$ (SD 36.12)** against **20.44 (61.63)** outdoors, higher in winter (20.17) than summer (11.19) and by day than night, with hourly patterns that contrast rather than track outdoors. Infiltration factors from median indoor/outdoor ratios of sulfur, nickel and iron were **51–70%**. Source contrasts were large: pets 24.09 vs 13.35, humidifiers 19.19 vs 13.16, smokers 16.76 vs 11.79, cooking 15.14 vs 10.96 $\mu\text{g}/\text{m}^3$. Every standard deviation exceeds its mean, so the distribution is episodic and a seasonal mean is a poor summary; five days per season is thin support for a long-term claim and no health estimate is reported yet. The transferable piece is the non-contact protocol — instruments redeployed by postal mail, which is how an unattended network is actually serviced.

doi: 10.1186/s12940-026-01334-3

Yu et al. 2026 Atmos Chem Phys*Continental pollutants modulate organic nitrogen and light absorption of marine organic aerosols*

Shipboard sampling over the Yellow and Bohai Seas across three seasons. Aerosol organic nitrogen averaged **0.35 $\pm$ 0.25 $\mu\text{g N m}^{-3}$** , 4–60% of total nitrogen, highest in autumn. Anthropogenic secondary pollutants — aged biomass burning and secondary nitrate — supplied **36–76% of ON**, 46–83% of the water-soluble and 39–89% of the water-insoluble fraction; in spring dust accounted for 55% of ON and above 80% during a dust episode, while summer carried a marine biogenic component. Organic-aerosol light absorption rose with both ON concentration and ON/OC ratio, so absorption capability per unit carbon increased with nitrogen enrichment rather than tracking mass. Three cruises give no annual coverage and apportionment is receptor modelling on a small campaign. The sensing consequence is direct: an optical instrument calibrated on one absorption regime is misapplied when composition shifts.

doi: 10.5194/acp-26-13175-2026

Sunyer-Caldú 2026 Environ Sci Technol*Beyond blood and urine: decoding the human chemical exposome across matrices and time*

A Perspective arguing that high-resolution mass spectrometry can profile thousands of chemical features per sample while study design has not kept pace — no investigation has combined external source monitoring, multi-matrix internal biomonitoring and toxicokinetic normalisation in one longitudinal framework. Three elements are proposed: multi-matrix sampling extending beyond blood and urine to accumulation archives, developmental matrices and external media; a Unified Exposure Scale placing heterogeneous matrices on a common toxicokinetically normalised dosimetric axis; and a two-track hazard-prioritisation pipeline. No data, no validation, and the Unified Exposure Scale is circular where it matters — it needs toxicokinetic parameters that do not exist for the molecular dark matter it is meant to rank. Logged for how air monitoring is being positioned inside the Global Human Exposome Project.

doi: 10.1021/acs.est.6c06509

Godri Pollitt 2026 — metadata only Environ Sci Technol*Beyond inhalation: air as an integrative matrix for exposome research*

ACS deposited this DOI on 18 September without an abstract and no text was recoverable from Crossref, Europe PMC or PubMed on per-DOI retry. Carried because it is the **second air-as-exposome-matrix Perspective in this same issue**, alongside Sunyer-Caldú, which makes the pair an editorial signal rather than a coincidence: air sampling is being argued into the exposome programme as an integrative matrix rather than merely an inhalation route. No argument or claim is asserted beyond the title. On the retry list.

doi: 10.1021/acs.est.6c05073

Sun et al. 2026 Environ Sci Technol*Nonlinear nitrate formation governed by aerosol water and VOC-sensitive oxidation in a humid urban basin*

Wintertime field observations in a humid, topographically confined Chinese urban basin, combined with a box model and thermodynamic calculations. Nitrate became the dominant PM_{2.5} component during pollution episodes, co-varying with NO₂, O_x and aerosol liquid water content. Pathway-resolved modelling split formation almost evenly between daytime OH+NO₂ (52%) and nighttime heterogeneous N₂O₅ hydrolysis (46%), the latter rising sharply during episodes. Binning regressions located thresholds at **RH 87%** and **ALWC 76.5 $\mu\text{g}/\text{m}^3$** above which formation efficiency rose abruptly; VOCs exerted slightly stronger control than NO_x, with oxygenated VOCs and alkenes dominant. One site, one winter, one box model — the thresholds are not validated out of sample. The RH threshold is directly load-bearing for sensor correction: it is the same humidity range in which optical low-cost sensors diverge from reference monitors.

doi: 10.1021/acs.est.6c08621

Zhao et al. 2026 Environ Sci Technol*Unveiling the importance of organic hydroxy acids-related autooxidation pathway in isoprene secondary organic aerosol formation*

Year-long PM_{2.5} measurements at **nine Pearl River Delta sites** quantifying six isoprene-derived highly oxygenated molecules (iHOMs). Annual mean **2.69 $\pm$ 3.52 ng/m^3** , higher in summer and autumn, dominated by methyltartaric acids (71%). Particle-phase tracer shares for the HOM, HMML and IEPOX pathways were 10%, 11% and **79%**; box-modelled gas-phase intermediates gave 25%, 22% and 53%, so IEPOX dominates isoprene SOA in the polluted PRD with the other two non-negligible in winter. Pathway partitioning tracked the ISOPO₂ isomer split. The standard deviation exceeds the mean, so the annual average describes episodes rather than a typical day, and tracer-to-SOA scaling relies on chamber yields measured well above ambient. These are **nanogram-level organics** — the composition layer no optical network sees.

doi: 10.1021/acs.est.6c06525

Frueh et al. 2026 J Expo Sci Environ Epidemiol*Source apportionment of volatile organic compounds in a Philadelphia fenceline community: results from THRIVEair*

Year-long community-designed VOC campaign in South Philadelphia run with Philly Thrive, collecting one-week integrated passive thermal-desorption samples of **38 VOCs at 20 sites**, July 2023 – June 2024, around a former oil refinery under redevelopment. Concentrations were higher in industrial than non-industrial and high- than low-traffic areas. **Benzene, the community's declared concern, was low overall** (median 1.21 $\mu\text{g}/\text{m}^3$, range 0.31–9.05), while (*m,p*)-xylenes emerged as the industrial-area pollutant of concern; toluene:benzene and xylene:ethylbenzene ratios were higher at industrial and high-traffic sites, indicating fresher emissions, and PMF apportioned sources. **One-week integration resolves none of the short plumes fenceline residents actually report**. The design is the template distributed PM networks should copy: community-set siting and a declared pollutant of concern rather than compliance checking.

doi: 10.1038/s41370-026-00976-2

Konon et al. 2026 — metadata only Atmos Environ

Seasonal variation in volatile organic compound emissions and secondary organic aerosol formation from unconventional oil and gas development

Same-day Elsevier deposit; nothing in Crossref, Europe PMC core, Europe PMC fullTextXML or the OpenAlex per-work record. The title indicates a seasonally resolved measurement campaign of VOC emissions from unconventional oil and gas development and the secondary organic aerosol those emissions form. In scope as a precursor study: **SOA is the fraction of basin PM_{2.5} that no primary-particle emission inventory captures**, and seasonality in the precursor flux propagates straight into any exposure model built on annual-average emission factors. Carried forward for pickup.

doi: 10.1016/j.atmosenv.2026.122368

Dou et al. 2026 — metadata only Environ Pollut

Volatile organic compounds in Chengdu: functional zone disparities and four-year dynamics in a basin city of southwest China

Same-day Elsevier deposit; no abstract in Crossref, Europe PMC core, Europe PMC fullTextXML or the OpenAlex per-work record. The title indicates a **four-year** ambient VOC record for Chengdu resolved by urban functional zone — the same Sichuan Basin whose humidity-driven nitrate chemistry Sun et al. model in this issue. In scope as precursor monitoring: functional-zone contrasts are the spatial scale distributed networks are built to resolve, and a four-year baseline is long enough to separate trend from meteorological variability. Carried forward for pickup.

doi: 10.1016/j.envpol.2026.129164

Burden, policy & mitigation**1 record****Massimi et al. 2026** Environ Sci Technol

Urban green walls under traffic stress: elemental composition of PM deposition and plant response

A nine-species green wall at a high-traffic urban site, followed for a year. PCA on the insoluble fraction of Al, Cr, Cs, Fe, Mn, Mo, Rb, Sb, Sn, Ti and Zn identified *Photinia × fraseri* as the highest and most stable retainer of non-exhaust traffic particles and *Pteris vittata* as the lowest, with SEM-EDS showing **up to 100-fold higher retention** in the former. The methodological finding travels furthest: **only insoluble elements traced deposition reliably**, because soluble and whole-leaf fractions are dominated by endogenous pools of nutrients and of tracers such as Cs and Rb, so conventional whole-leaf digestion overestimates deposition. *Photinia* showed marked seasonal physiological and transcriptional modulation; *Pteris* did not. One wall at one site with no paired ambient PM mass measurement, so species are ranked against each other but no removal flux is quantified and no air-quality benefit can be inferred.

doi: 10.1021/acs.est.6c10086

Corpus register

First author	Focus	Journal	Design	Metric	Region
Neuro / mental health					
Morrel et al.	Outdoor Air Pollution, Perivascular Space Morphology, and Cognition in Preadolescence.	<i>Biol Psychiatry Glob Open Sci</i>	Cross-sectional cohort imaging + weighted quantile sum regression	Annual-average PM _{2.5} , O ₃ , NO ₂ and 15 PM _{2.5} components, spatiotemporal models at residence	USA
Temporao et al.	Daily associations between outdoor air pollution and sleep quality.	<i>medRxiv (preprint)</i>	Panel study with ambulatory EEG	Daytime outdoor PM _{2.5} , NO ₂ and O ₃ aggregated from city monitoring stations; same-day and previous-day	Hungary
Cardiovascular & metabolic					
Chiang et al.	Ambient Carbon Monoxide as a Delayed Trigger of Acute Aortic Dissection: A Time-Stratified Case-Crossover Study in a Tropical Industrial Metropolis.	<i>J Am Heart Assoc</i>	Case-crossover	PM _{2.5} , PM ₁₀ , CO, NO ₂ , O ₃ , SO ₂ from 24 fixed stations; lags 0-3 d, per 1 ppm and per IQR	Taiwan
Fleming & Fleming	Tire Wear Pollution and Cardiovascular Risk.	<i>Am J Med</i>	Metadata only (no abstract)	Tire-wear (non-exhaust) particles; no exposure metric retrievable	Global
Zhang et al.	Cardiopulmonary hospitalization risks from wildfire-specific and non-wildfire PM _{2.5} in 20 US states.	<i>Nat Commun</i>	Self-controlled design	Two-year average wildfire-specific vs non-wildfire PM _{2.5} , modelled and source-separated, per 1 µg/m ³	USA
Respiratory & allergic					
Chen et al.	Residential Green and Blue Space, the Natural Environment, Genetic Susceptibility, and Incident COPD.	<i>Int J Chron Obstruct Pulmon Dis</i>	Prospective cohort	Residential green space, blue space and natural environment at 300 m and 1000 m buffers; no PM term reported	UK
San Juan-Reyes et al.	Short-term effects of ambient air pollution exposure on respiratory hospitalizations among children in the State of Mexico.	<i>Environ Sci Pollut Res Int</i>	Case-crossover	Daily PM _{2.5} , PM ₁₀ , CO, NO ₂ and O ₃ from fixed monitoring stations; primary window lag 1-7 cumulative	Mexico
Mechanistic toxicology					
Dulong et al.	Effects of fine particulate matter on neuroblastoma cell line: metabolic and lipid deregulations and their consequences on the onset of neurological diseases.	<i>Biochimie</i>	Narrative review	PM _{2.5} , in vitro dosing; no harmonised mass concentration or duration reported	Global
Gao et al.	Integrative transcriptomic analysis identifies CCL22-associated immune signatures in air pollution-related atopic dermatitis.	<i>Front Public Health</i>	Integrative transcriptomics + murine validation	No measured PM; pollutant-associated gene sets taken from a toxicogenomic database and intersected with atopic-dermatitis transcriptomes	Not stated
Kou et al.	Pollutants regulate changes in pathological markers of neurodegenerative diseases: a new perspective in environmental toxicology.	<i>Front Toxicol</i>	Narrative review	Air pollutants treated as one class alongside metals and POPs; no exposure metric	Global

First author	Focus	Journal	Design	Metric	Region
Vera-Colon et al.	Exposure to airborne fine particulate matter disrupts human embryonic stem cell osteogenic differentiation.	<i>Part Fibre Toxicol</i>	Metadata only (no abstract)	PM composition, dose and collection route not recoverable from metadata	Not stated
Wu et al.	Beta-carotene alleviates PM _{2.5} -induced pulmonary fibrosis via suppression of the Frizzled5-mediated Wnt5a/Ca2+ pathway.	<i>Ecotoxicol Environ Saf</i>	Animal exposure study + in vitro receptor overexpression	Intratracheal PM _{2.5} instillation; dose, source and treatment concentration not reported	China
Occupational & indoor					
Assan et al.	Evaluation of Ambient Air Particulate Matter, Urine Arsenic Levels and the Effect on Residents Health in an Artisanal Mining Community and an Incorporated Mining Community in Ghana.	<i>Health Sci Rep</i>	Cross-sectional (comparative community)	Ambient PM _{2.5} and PM ₁₀ by Modular Air Quality System; urinary arsenic by atomic absorption spectrophotometry	Ghana
Ikeda et al.	Characterizing layout-dependent breathing-zone contaminant concentration in office spaces with mixing ventilation using an occupant-to-exhaust distance metric.	<i>Build Environ</i>	Metadata only (no abstract)	Breathing-zone contaminant concentration under mixing ventilation; no quantities retrievable	Japan
Khan et al.	Exposure to cement industry emissions and respiratory health outcomes: a global systematic review and meta-analysis from 2000 to 2025.	<i>Front Public Health</i>	Systematic review + random-effects meta-analysis	Cement dust and cement-plant emissions, occupational and residential; no harmonised gravimetric metric across studies	Global / multi-region
Sensing, forecasting & instrumentation					
Bai et al.	Role of low-level jet evolution in vertical aerosol redistribution: a Doppler wind lidar study over East China.	<i>Atmos Meas Tech</i>	Remote sensing / lidar retrieval	Coherent Doppler wind lidar backscatter, spectral width and shear profiles against surface PM ₁₀	China
Huang et al.	Interpretable Spatiotemporal Coupling Network for Robust Emission Prediction and Implicit Mechanism Mining in Sludge Gasification.	<i>Environ Sci Technol</i>	Deep-learning forecasting	Continuous industrial stack monitoring of PM, SO ₂ and NO _x from a sludge-gasification plant, with feature-level optimal time lags	Not stated
Kalluri et al.	Long-term (2008-2024) analysis of black carbon aerosols over a semi-arid rain-shadow region in southern peninsular India: trends, wavelet analysis, machine learning approaches and source regions.	<i>Atmos Environ</i>	Metadata only (no abstract)	Black carbon, presumably aethalometer-derived; instrument, wavelength set and trend magnitudes not recoverable	India
Krajnc Pogacnik et al.	Selective real-time monitoring of respiratory droplets for indoor air safety in educational environments.	<i>Build Environ</i>	Metadata only (no abstract)	Respiratory-droplet detection; sensing principle, size range and selectivity mechanism not recoverable	Not stated
Raju et al.	Investigations of black carbon aerosol sources and strengths over a rural-valley location in Haryana, India.	<i>J Environ Sci Health A</i>	Multi-year aethalometer monitoring + source apportionment	Equivalent black carbon, seven-wavelength AE33, 880 nm; annual and seasonal means in $\mu\text{g}/\text{m}^3$	India

First author	Focus	Journal	Design	Metric	Region
Exposure assessment & modelling					
Dou et al.	Volatile organic compounds in Chengdu: Functional zone disparities and four-year dynamics in a basin city of Southwest China.	<i>Environ Pollut</i>	Metadata only (no abstract)	Ambient VOCs by functional zone, four-year record; no quantities retrievable	China
Frueh et al.	Source apportionment of volatile organic compounds in a Philadelphia fenceline community: results from THRIVEair.	<i>J Expo Sci Environ Epidemiol</i>	Community passive-sampler campaign + PMF source apportionment	38 VOCs, one-week integrated passive thermal-desorption tubes, 20 sites, 12 months	USA
Godri Pollitt	Beyond Inhalation: Air as an Integrative Matrix for Exposome Research.	<i>Environ Sci Technol</i>	Metadata only (no abstract)	No measured PM; air framed as an exposome sampling matrix	Global / multi-region
Kim et al.	Residential indoor air monitoring to assess long-term exposure to air pollution in susceptible populations for epidemiology: overview and early findings.	<i>Environ Health</i>	Long-term indoor monitoring campaign	Indoor and matched outdoor PM _{2.5} plus 36 chemical components, RTI MicroPEM, 5-day runs in each of four seasons, hourly resolution	South Korea
Konon et al.	Seasonal Variation in Volatile Organic Compound Emissions and Secondary Organic Aerosol Formation from Unconventional Oil and Gas Development.	<i>Atmos Environ</i>	Metadata only (no abstract)	VOC emissions and SOA formation potential; no quantities retrievable	USA
Sun et al.	Nonlinear Nitrate Formation Governed by Aerosol Water and VOC-Sensitive Oxidation in a Humid Urban Basin.	<i>Environ Sci Technol</i>	Field observations + box model + thermodynamic equilibrium	Wintertime PM _{2.5} composition, NO ₂ , O ₃ and aerosol liquid water content	China
Sunyer-Caldu	Beyond Blood and Urine: Decoding the Human Chemical Exposome across Matrices and Time.	<i>Environ Sci Technol</i>	Perspective / framework proposal	No measured PM; air positioned as one external matrix within a multi-matrix exposome design	Global / multi-region
Yu et al.	Continental pollutants modulate organic nitrogen and light absorption of marine organic aerosols over East Asian marginal seas.	<i>Atmos Chem Phys</i>	Field campaign / shipborne	Shipboard aerosol organic nitrogen, water-soluble and water-insoluble fractions, ON/OC ratio and light absorption over three seasons	China
Zhao et al.	Unveiling the Importance of Organic Hydroxy Acids-Related Autoxidation Pathway in Isoprene Secondary Organic Aerosol Formation.	<i>Environ Sci Technol</i>	Year-long regional tracer measurements + box model	PM _{2.5} -phase isoprene SOA tracers, six iHOM species, ng/m ³ , nine sites	China
Burden, policy & mitigation					
Massimi et al.	Urban Green Walls under Traffic Stress: Linking the Elemental Composition of Particulate Matter Deposition to Plant Physiological and Gene Expression Responses.	<i>Environ Sci Technol</i>	Biomonitoring + source apportionment	Multi-elemental fractionation of PM deposited on green-wall foliage - insoluble, soluble and whole-leaf fractions - with SEM-EDS, over one year	Italy

Provenance and method

Entry window **18 September 2026**, single day, continuous with the previous issue. This issue was built in **two passes**: a 10:33 CDT pass that saw roughly ten hours of the window and admitted 17 records, and a 23:0x re-harvest of the same window that admitted 13 more. The second pass is what makes the window complete; see the run note above. Legs run in the second pass: **PubMed** E-utilities on the health and instrumentation axes as separate queries (**12** health, 0 sensing); **Europe PMC** `CREATION_DATE` (**46**); **Crossref** by `ISSN` (**53**, again the highest-yield leg). **OpenAlex** returned HTTP 429 and **arXiv** HTTP 406, both known-broken legs. The **PubMed connector** added **no** net on-topic record: of its twelve PMIDs, three were unseen on the PMID key, but two of those three are records this issue already carries — the morning pass admitted them from Europe PMC with no PMID attached, so only the DOI key caught them. The one genuinely new record was an x-ray air-kerma Monte Carlo study, rejected as off-topic. **Nine** Europe PMC records were dropped at the source as retro-index artefacts (AGRICOLA and PMC records returned inside the window whose first publication dates precede the issue month — the gate hand-applied on 18 September now lives in the screen script). The remaining 102 raw records deduplicated to 97, of which 25 were already carried, leaving 72 fresh candidates of which **13 were admitted**. Across both passes, **132 fresh candidates were screened and 30 admitted**. **Consensus** returned 10 sensor-calibration papers and admitted **none** for the third consecutive issue: 5 matched archive titles and all 5 survivors failed a Crossref created check (2026-01-02 to 2026-06-15), including a 1,013-sensor global PurpleAir calibration whose deposit date is 15 June. A **new abstract-recovery route was tested and failed**: the free OpenAlex per-work endpoint returns an abstract only where one was deposited to Crossref, so it recovers nothing from the publishers that withhold them. **Eight records are carried on metadata alone** and assert no finding. All 30 DOIs were resolved against Crossref before build (0 fail, 0 warn). Subtopic, design, particle-metric and geography labels are assigned from title, abstract, keywords and MeSH; assignment is single-label by dominant contribution and is a judgement, not a controlled vocabulary. Effect estimates are transcribed verbatim and use heterogeneous exposure contrasts — see the forest-plot caption. Abstracts remain the copyright of their respective publishers.